

Hsin-Ying Lee

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EDUCATION

- **University of California, Merced** Merced
Ph.D. in Electrical Engineering and Computer Science Aug 2024 – Present
 - **Image and Video Generation:** advised by Prof. Ming-Hsuan Yang
 - **GPA:** 4.0/4.0
- **National Taiwan University** Taipei
Master's in Computer Science and Information Engineering Sep 2020 – Nov 2022
 - **Video Question Answering:** advised by Prof. Winston H. Hsu
 - **GPA:** 4.24/4.3
 - **Rank:** 9/136
- **National Taiwan University** Taipei
Bachelor's in Information Management Sep 2016 – Jun 2020
 - **GPA:** 4.01/4.3
 - **Rank:** 1/42

EXPERIENCE

- **Adobe Research** San Jose
Research Scientist Intern Jul 2025 - Present
 - **Mentor:** Zhixin Shu
 - **Motion-Controlled Video Generation with Visual Reasoning:** Integrate visual reasoning to generate physically plausible and common-sense consistent motion for video generation.
- **Vision and Learning Lab** Merced
Visiting Student Jul 2023 - Jan 2024
 - **Mentors:** Hsin-Ying (James) Lee, Hung-Yu Tseng
 - **Generation Prior for Dense Prediction:** Leverage image generative prior for dense prediction tasks such as depth and normal prediction. The method shows strong generalizability.
- **Mobile Drive Technology** Taipei
Research Engineer Intern Sep 2022 - Oct 2022
 - **Lane Detection for Autonomous Driving:** Adapt and implement lane detection for edge devices and local environments
- **ASUS Intelligent Cloud Services** Taipei
Software Engineer Intern Jul 2019 - Aug 2019
 - **Anti-Spoofing for Face Unlock:** Develop anti-spoofing by predicting face depth for face unlock on mobile phones

PUBLICATIONS

Published under Lee Hsin-Ying from 2025

- **MotiMotion: Motion-Controlled Video Generation with Visual Reasoning**
Lee Hsin-Ying, Hanwen Jiang, Yiqun Mei, Jing Shi, Ming-Hsuan Yang, Zhixin Shu
International Conference on Machine Learning (ICML) 2026 [[Paper](#)] [[Webpage](#)]
- **Consistent Subject Generation via Contrastive Instantiated Concepts**
Lee Hsin-Ying, Kelvin C.K. Chan, Ming-Hsuan Yang
Transactions on Machine Learning Research (TMLR) 2025 [[Paper](#)] [[Webpage](#)]

- **Exploiting Diffusion Prior for Generalizable Dense Prediction**
Hsin-Ying Lee, Hung-Yu Tseng, Hsin-Ying Lee, Ming-Hsuan Yang
Computer Vision and Pattern Recognition (CVPR) 2024 [[Paper](#)] [[Webpage](#)]
- **Learning Fine-grained Visual Understanding for Video Question Answering via Decoupling Spatial-Temporal Modeling**
Hsin-Ying Lee, Hung-Ting Su, Bing-Chen Tsai, Tsung-Han Wu, Jia-Fong Yeh, Winston H. Hsu
The British Machine Vision Conference (BMVC) 2022 **Spotlight** [[Paper](#)] [[Webpage](#)]
- **D2ADA: Dynamic Density-aware Active Domain Adaptation for Semantic Segmentation**
Tsung-Han Wu, Yi-Syuan Liou, Shao-Ji Yuan, **Hsin-Ying Lee**, Tung-I Chen, Kuan-Chih Huang, Winston H. Hsu
European Conference on Computer Vision (ECCV) 2022
- **ReDAL: Region-based and Diversity-aware Active Learning for Point Cloud Semantic Segmentation**
Tsung-Han Wu, Yueh-Cheng Liu, Yu-Kai Huang, **Hsin-Ying Lee**, Hung-Ting Su, Ping-Chia Huang, Winston H. Hsu
International Conference on Computer Vision (ICCV) 2021
- **CrossDTR: Cross-view and Depth-guided Transformers for 3D Object Detection**
Ching-Yu Tseng, Yi-Rong Chen, **Hsin-Ying Lee**, Tsung-Han Wu, Wen-chin Chen, Winston H. Hsu
International Conference on Robotics and Automation (ICRA) 2023
- **Coarse-To-Fine Point Cloud Registration with SE(3)-Equivariant Representations**
Cheng-Wei Lin, Tung-I Chen, **Hsin-Ying Lee**, Wen-chin Chen, Winston H. Hsu
International Conference on Robotics and Automation (ICRA) 2023
- **CFVS: Coarse-to-Fine Visual Servoing for 6-DoF Object-Agnostic Peg-In-Hole Assembly**
Bo-Siang Lu, Tung-I Chen, **Hsin-Ying Lee**, Winston H. Hsu
International Conference on Robotics and Automation (ICRA) 2023

TEACHING

- **Teaching Assistant** UC Merced
 - CSE185 Introduction to Computer Vision: Fall 2025
 - CSE22 Introduction to Programming: Fall 2024, Spring 2025
- **Teaching Assistant** National Taiwan University
 - Statistics: Spring 2019

SERVICES

- **Conference Reviewer:** Computer Vision and Pattern Recognition (CVPR), European Conference on Computer Vision (ECCV), Neural Information Processing Systems (NeurIPS)
- **Journal Reviewer:** International Journal of Computer Vision (IJCV), Computer & Graphics

AWARDS

- **Dean's List**
 - top 5% academic standing of the department in the semester Spring 2018, Fall 2018, Spring 2019, Fall 2019